

FF-SOFT Heatmap — Single Instance Analysis

Dataset: OpenBookQA | Model: LLaMA-3.2-3B-Instruct | Instance: 9-368

Question: Birds carrying away fruit helps the tree

A) A): grow

B) B): fertilize [Predicted]

C) C): reproduce [Correct]

D) D): conquer

Initial no-CoT prediction: B) B): fertilize (probability = 0.6863)

Reasoning Steps:

Step 1: Δp = +0.046 (supports prediction)

- Birds carrying away fruit helps the tree in a specific way.

Step 2: Δp = +0.036 (supports prediction)

- The correct answer should be related to the benefit the tree receives from this action.

Step 3: Δp = +0.676 (supports prediction)

- Fertilization is not directly related to birds carrying away fruit.

Step 4: Δp = -0.305 (counteracts prediction)

- Reproduction is not directly related to birds carrying away fruit.

Step 5: Δp = -0.288 (counteracts prediction)

- Conquering is not a biological process related to the interaction between birds and trees.

Step 6: Δp = +0.621 (supports prediction)

- The most direct benefit the tree receives from birds carrying away fruit is through the process of fertilization, a...

